Finite-Tick Corrections for Concentrated Liquidity: Non-Universality, Misalignment, and Optimal Range Design

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Abstract

Echenim, Gobet, and Maurice (2025) derived asymptotic formulas for liquidity provider (LP) fee revenue in Uniswap v3, proving convergence to an integral of the price local time as the tick spacing $\gamma \to 0$. In practice, γ is fixed by the pool's fee tier and can be as large as 2%. We derive three sets of results on the finite-tick regime. **(1) Non-universality of κ_1.** Under a general constant-drift diffusion $dX_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, the first-order correction coefficient is $\kappa_1 = 1/2 + \mu/(2\sigma^2)$, which equals $1/4$ *only* for risk-neutral geometric Brownian motion ($\mu = -\sigma^2/2$). Under the physical measure with expected return $r > 0$, the correction is $\kappa_1^P = 1/4 + r/(2\sigma^2)$, which for typical crypto parameters is 2–3 times the risk-neutral value. The result decomposes into a prefactor contribution and an Euler–Maclaurin boundary term, and the cancellation $f'(0) = 0$ that produces "universality" is a precise algebraic artifact of the risk-neutral GBM drift. **(2) Misaligned-range correction.** When the LP range boundary falls between two tick levels with fractional offset δ, the boundary tick earns a fraction $R(\delta) = 2(1-\delta)/(2-\delta)$ of the interior tick's fee. Unlike the classical Broadie–Glasserman–Kou correction for temporally discretized barriers, this spatial correction is a rational function of δ and does not involve the Wiener–Hopf constant $\zeta(1/2)/\sqrt{2\pi}$, because the continuous price process crosses every tick level with probability one. **(3) Optimal range width.** The finite-tick correction shifts the fee-maximizing range width downward by 1–2 ticks relative to the EGM asymptotic, with the largest effect in the 0.05% pool. We validate all results against semi-analytical computations and Monte Carlo simulations.

1 Introduction

1.1 Concentrated liquidity in automated market makers

Automated Market Makers (AMMs) enable decentralized exchange of crypto-assets without a traditional order book. The constant-product formula $x \cdot y = k$, introduced by Uniswap v1 [Adams et al., 2021], requires liquidity providers (LPs) to spread capital across the entire price range $[0, \infty)$, resulting in low capital efficiency. Uniswap v3 [Adams et al., 2021] addressed this through *concentrated liquidity*: LPs restrict their capital to a chosen price range $[p_\ell, p_u]$, earning swap fees only when the market price falls within this range. The concentration amplifies fee income per unit of capital but exposes the LP to *impermanent loss* (IL) when the price exits the range.

The tick mechanism discretizes the price space into levels spaced multiplicatively by $(1 + \gamma)$, where γ depends on the pool's fee tier:
The tick spacing is coarser for higher fee tiers, reflecting the intuition that volatile pairs require wider spreads. Uniswap v4 retains the same tick mechanism with identical fee tier structure, so all results in this paper apply equally to both versions.

Table 1: Uniswap v3 fee tiers and tick spacings.

Fee tier ϕ	Tick spacing δ_π	γ (approx.)	Typical use case
0.01%	1	0.01%	Stablecoins
0.05%	10	0.10%	Correlated pairs
0.30%	60	0.60%	Most pairs
1.00%	200	2.02%	Exotic/volatile

The discreteness of the tick grid introduces a systematic gap between the continuous-limit fee formulas used for LP strategy optimization and the actual fee revenue earned on-chain. Quantifying this gap is the central problem of this paper.

1.2 The EGM asymptotic framework

Echenim et al. [2025] established a rigorous asymptotic framework for LP fee revenue. Their main result (Theorem 4.1) shows that as $\gamma \to 0$, the renormalized fee revenue $\gamma \cdot \text{Fees}$ converges in probability to an integral involving the local time of the log-price process:

$$\gamma \cdot \text{Fees} \xrightarrow{\mathbb{P}} \frac{\phi}{1-\phi} \int_{p_\ell}^{p_u} A_T^b(p)\, b^{-3/2}\, \mathrm{d}b, \tag{1}$$

where $A_T^b(p)$ is the occupation functional of the price process at level b, which under continuous diffusion models reduces to the semimartingale local time $L_T^{\log b}(X)/b$ of the log-price. Under geometric Brownian motion (GBM), they obtained an explicit expected-fee formula (Theorem 4.5). These results hold in the limit $\gamma \to 0$; the finite-γ regime was left unexplored.

The practical relevance of finite-γ corrections is immediate: for the 1% fee tier, $\gamma \approx 0.02$, and for a 20-tick LP range, the correction to the EGM asymptotic can exceed 1.5% of total fee revenue. For an LP deploying \$10M of capital, this corresponds to approximately \$15,000 per year in systematic estimation error.

1.3 Our contributions

We derive three sets of results that substantially extend the EGM framework to the finite-tick regime.

Contribution 1: Non-universality of κ_1. Under any constant-drift diffusion $dX_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, we prove that the first-order correction coefficient is

$$\kappa_1(\mu, \sigma) = \frac{1}{2} + \frac{\mu}{2\sigma^2}, \tag{2}$$

so that $\Psi(\gamma)/F_0^Y(\gamma) = 1 + \kappa_1\gamma + O(\gamma^2)$. The value $\kappa_1 = 1/4$ holds *only* for the risk-neutral GBM drift $\mu = -\sigma^2/2$. Under the physical measure with expected excess return $r > 0$, the correction becomes $\kappa_1^P = 1/4 + r/(2\sigma^2)$, which for Bitcoin ($r \approx 0.5$, $\sigma \approx 0.8$) gives $\kappa_1^P \approx 0.64$—more than 2.5 times the risk-neutral value. The result decomposes into a prefactor term (from the nonlinear expansion of the fee-per-crossing formula) and an Euler–Maclaurin boundary term (from the derivative $f'(0)$ of the weight function at the origin). The precise cancellation $f'(0) = 0$ that produces "universality" is shown to be an algebraic artifact of the risk-neutral GBM drift.

Contribution 2: Misaligned-range correction. When the LP range boundary does not align with the tick grid, the boundary tick earns a reduced fraction of the interior tick's fee. We derive the explicit formula

$$R(\delta) = \frac{2(1-\delta)}{2-\delta}$$

for the boundary tick efficiency ratio, where $\delta \in [0, 1)$ is the fractional offset. This correction is a smooth rational function that interpolates from $R(0) = 1$ (aligned) to $R(1^-) = 0$ (boundary at the tick level). We prove that, unlike the classical Broadie–Glasserman–Kou (BGK) correction for temporally discretized barriers, the spatial misalignment correction does *not* involve the Wiener–Hopf constant $\beta = \zeta(1/2)/\sqrt{2\pi}$. The fundamental reason is that a continuous price process crosses every spatial level with probability one (no overshoot), whereas discrete temporal monitoring allows the process to jump past a barrier between observation dates.

Contribution 3: Optimal range width. We derive the fee-maximizing range width $N^*(\gamma)$ under the corrected fee formula and show that the second-order correction $-N^2\gamma^2/6$ acts as an additional penalty on wide ranges, shifting the optimum downward by 1–2 ticks relative to the EGM asymptotic. The correction is most significant for the 0.05% pool ($\gamma = 0.001$) where optimal ranges are widest. For a \$1M LP position in this pool, the corrected range design improves daily fee revenue by approximately \$200.

1.4 Mathematical structure: two sources of correction

The finite-tick correction has a clean two-source decomposition that illuminates its mathematical structure:

(i) **Prefactor correction** κ_1^{ψ}: The fee per crossing involves the nonlinear function $c(h) = (\gamma/h)(e^{h/2} - 1) \cdot S(h)/S(2h)$, where S is the scale function. Taylor expanding $c(h)$ around $h = 0$ gives a first-order contribution $\kappa_1^{\psi} = 3/4 + \mu/\sigma^2$.

(ii) **Euler–Maclaurin correction** κ_1^{EM}: The discrete sum $\sum hf(kh)$ over the tick grid approximates the integral $\int f(x)\,\mathrm{d}x$ with an error governed by the Euler–Maclaurin formula. When $f'(0) \neq 0$, the leading boundary term contributes $\kappa_1^{EM} = -1/4 - \mu/(2\sigma^2)$.

The total correction is $\kappa_1 = \kappa_1^{\psi} + \kappa_1^{EM} = 1/2 + \mu/(2\sigma^2)$. Under risk-neutral GBM ($\mu = -\sigma^2/2$), the Euler–Maclaurin contribution simplifies to $\kappa_1^{EM} = 0$ (because $f'(0) = 0$), leaving only $\kappa_1^{\psi} = 1/4$. This cancellation is *specific* to the risk-neutral drift; any other drift breaks it.

1.5 Connection to discrete barrier corrections

The classical result of Broadie et al. [1997, 1999] shows that a discretely monitored barrier option price can be corrected by shifting the barrier by $\beta\sigma\sqrt{\Delta t}$, where $\beta = \zeta(1/2)/\sqrt{2\pi} \approx -0.5826$ arises from the Wiener–Hopf factorization of the random walk maximum's distribution. Hörfelt [2003] extended this to second order, and Howison and Steinberg [2007] provided an alternative derivation via matched asymptotic expansions. Feng and Linetsky [2008] further extended the framework to exponential Lévy processes using Hilbert transforms.

In our setting, the tick grid forms a *spatial* discretization, not a temporal one. The price process is continuous in time and crosses each tick level with probability one—there is no random walk overshoot, so the Spitzer–Baxter machinery underlying BGK does not apply. Our corrections are instead algebraic (from the fee formula expansion) and analytic (from the Euler–Maclaurin formula

applied to the crossing-count sum). For misaligned range boundaries, the correction is a rational function of the fractional offset, not an irrational constant involving $\zeta(1/2)$.

This distinction between spatial and temporal discretization corrections appears to be new in the AMM literature, though it is implicit in the matched asymptotics framework of Howison and Steinberg [2007].

1.6 Related work

The qualitative connection between AMM positions and barrier options was noted by Paone [2025]. The technical backbone of the EGM proof builds on the local time approximation results of Gobet and Landon [2014] and Gobet and Stazhynski [2020]. Clark and Hassani [2023] analyzed LP returns under GBM using option pricing techniques. Fukasawa et al. [2023] studied optimal concentrated liquidity provision as a continuous-time stochastic control problem. Carta and Fabrizio [2024] derived fee revenue formulas for general semimartingale price processes using multiplicative Doob decompositions. To our knowledge, no prior work derives a finite-tick correction for AMM fee revenue, nor identifies the non-universality of the first-order correction coefficient.

1.7 Organization

Section 2 introduces the model setup and notation. Section 3 states and proves the general first-order correction formula for arbitrary constant drift. Section 4 provides a structural analysis of the $f'(0) = 0$ cancellation and its consequences. Section 5 specializes to GBM and derives the full two-term correction including the second-order range-dependent term. Section 6 treats misaligned range boundaries. Section 7 derives the optimal range width under the corrected formula. Section 8 presents numerical validation. Section 9 discusses economic implications and extensions. Section 10 concludes. Detailed calculations are collected in the Appendix.

2 Setup

2.1 Price dynamics and the log-price process

Let p_t denote the exchange rate (price of asset X in units of asset Y) at time $t \in [0, T]$. Under a general constant-drift diffusion model, the log-price $X_t = \log p_t$ satisfies

$$dX_t = \mu \, \mathrm{d}t + \sigma \, \mathrm{d}W_t, \qquad X_0 = x_0, \tag{3}$$

with constant drift $\mu \in \mathbb{R}$ and volatility $\sigma > 0$. This encompasses several important cases:

- **Risk-neutral GBM**: $\mu = -\sigma^2/2$, corresponding to $dp_t/p_t = \sigma \, \mathrm{d}W_t$ under $\mathbb{P}^*$.
- **Physical-measure GBM**: $\mu = r - \sigma^2/2$, where r is the expected (log-)return.
- **Arithmetic Brownian motion (ABM)**: $\mu = 0$, $X_t = x_0 + \sigma W_t$, modeling a driftless log-price.
- **General drift**: μ arbitrary, e.g., $\mu > 0$ for trending assets, $\mu < 0$ for deflationary tokens.

2.2 Tick grid and LP positions

The tick spacing parameter $\gamma > 0$ induces a log-price grid with spacing

$$h = \log(1 + \gamma) = \gamma - \frac{\gamma^2}{2} + \frac{\gamma^3}{3} - \cdots . \tag{4}$$

The grid levels are $\{kh : k \in \mathbb{Z}\}$. An LP deposits unit liquidity on N consecutive ticks centered at $x_0 = 0$ (without loss of generality), giving a range

$$R_\gamma = \Big[-\frac{Nh}{2},\ \frac{Nh}{2}\Big] \tag{5}$$

in log-price space. We assume N is even throughout.

2.3 Scale function and crossing probabilities

The scale function of the diffusion (3) is

$$S(x) = \int_0^x \exp\Big(-\frac{2\mu}{\sigma^2}\, y\Big)\, \mathrm{d}y = \frac{1}{\alpha}\big(e^{\alpha x} - 1\big), \tag{6}$$

where we define the *drift parameter*

$$\alpha \coloneqq -\frac{2\mu}{\sigma^2}. \tag{7}$$

For risk-neutral GBM, $\alpha = 1$; for ABM, $\alpha = 0$ (and $S(x) = x$ by continuity). The gambler's ruin formula gives the probability of an upcrossing from a grid level:

Lemma 2.1 (Crossing probability)**.** *Starting at grid level kh, the probability of reaching $(k{+}1)h$ before $(k{-}1)h$ under the diffusion* (3) *is*

$$p_\uparrow(h) = \frac{S(h)}{S(2h)} = \frac{e^{\alpha h} - 1}{e^{2\alpha h} - 1} = \frac{1}{1 + e^{\alpha h}}. \tag{8}$$

In particular:

$$p_\uparrow(h) = \frac{1}{2} - \frac{\alpha h}{4} + O(h^2), \tag{9}$$

so the upcrossing probability is less (resp. greater) than $1/2$ when $\alpha > 0$ (resp. $\alpha < 0$), i.e., when the drift is negative (resp. positive).

Proof. By the standard exit problem for Brownian motion with drift, the probability of reaching level a before level $-b$ starting from 0 is $S(b)/S(a+b)$ when S is the scale function. Setting $a = b = h$ and using the explicit form (6) gives

$$p_\uparrow = \frac{S(h)}{S(2h)} = \frac{e^{\alpha h} - 1}{e^{2\alpha h} - 1} = \frac{1}{1 + e^{\alpha h}}.$$

The last equality follows from the factorization $e^{2\alpha h}{-}1 = (e^{\alpha h}{-}1)(e^{\alpha h}{+}1)$. For the Taylor expansion, write $e^{\alpha h} = 1 + \alpha h + (\alpha h)^2/2 + O(h^3)$, so $1 + e^{\alpha h} = 2 + \alpha h + O(h^2)$, and $1/(2 + \alpha h + O(h^2)) = 1/2 - \alpha h/4 + O(h^2)$. □

2.4 Local time and occupation density

The semimartingale local time $L_T^x = L_T^x(X)$ of the process X at level x up to time T is defined via the occupation times formula:

$$\int_0^T g(X_s)\, \mathrm{d}\langle X\rangle_s = \int_{-\infty}^{\infty} g(x)\, L_T^x\, \mathrm{d}x \tag{10}$$

for all measurable g. Since $\langle X\rangle_t = \sigma^2 t$, this is equivalent to $\int_0^T g(X_s)\,\mathrm{d}s = \sigma^{-2}\int g(x)\,L_T^x\,\mathrm{d}x$.

The expected local time has the explicit representation

$$\mathbb{E}[L_T^x] = \sigma^2 \int_0^T p_X(t,0,x)\,\mathrm{d}t, \tag{11}$$

where $p_X(t,0,x)$ is the transition density of X:

$$p_X(t,0,x) = \frac{1}{\sigma\sqrt{2\pi t}}\exp\Big(-\frac{(x-\mu t)^2}{2\sigma^2 t}\Big). \tag{12}$$

2.5 Fee structure and the crossing-count representation

Following Echenim et al. [2025], the Y-token fee accumulated over $[0,T]$ from upcrossings at tick level kh is

$$\text{fee per upcrossing at } kh = \pi_k \cdot \big((1+\gamma)^{1/2} - 1\big) \cdot \frac{\phi}{1-\phi}, \tag{13}$$

where $\pi_k = e^{kh/2}$ is the square-root price at tick level kh. The factor $(1+\gamma)^{1/2} - 1$ is the relative change in the virtual reserve $\sqrt{p}$ across one tick.

Definition 2.2 (Prefactor function)**.** Define the crossing-count coefficient

$$c(h) \coloneqq \frac{\gamma}{h}\cdot\frac{e^{h/2}-1}{1+e^{\alpha h}}, \tag{14}$$

and the normalized prefactor

$$\psi(h) \coloneqq \frac{4c(h)}{h}. \tag{15}$$

The expected renormalized Y-fee is then

$$\Psi(\gamma) = \frac{\phi}{1-\phi}\sum_{k=-N/2}^{N/2-1} c(h)\cdot e^{kh/2}\cdot\mathbb{E}[L_T^{kh}]. \tag{16}$$

2.6 The EGM asymptotic formula

The Y-fee component of the EGM limit for the range R_γ is

$$F_0^Y(\gamma) = \frac{\phi}{4(1-\phi)}\int_{-Nh/2}^{Nh/2} e^{-3x/2}\,\mathbb{E}[L_T^x]\,\mathrm{d}x. \tag{17}$$

This is obtained from the change of variables $b = e^x$ applied to (1), using $A_T^b = L_T^{\log b}/b$ and $b^{-3/2}\,\mathrm{d}b = e^{-3x/2}\,\mathrm{d}x$.

Our goal is to compute the ratio $\Psi(\gamma)/F_0^Y(\gamma)$ as an expansion in powers of γ (equivalently, h).

3 The General First-Order Correction

We now state and prove our main structural result: the first-order correction coefficient for general constant drift.

3.1 Prefactor expansion

Lemma 3.1 (General prefactor expansion)**.** *For the prefactor $\psi(h)$ defined in* (15)*, with $\alpha = -2\mu/\sigma^2$:*

$$\psi(h) = 1 + h\,\kappa_1^{\psi} + O(h^2), \tag{18}$$

where

$$\kappa_1^{\psi} = \frac{3}{4} + \frac{\mu}{\sigma^2} = \frac{3}{4} - \frac{\alpha}{2}. \tag{19}$$

Proof. We expand each factor in $c(h)$ separately.

Step 1. The ratio γ/h: Since $\gamma = e^h - 1 = h + h^2/2 + h^3/6 + O(h^4)$,

$$\frac{\gamma}{h} = 1 + \frac{h}{2} + \frac{h^2}{6} + O(h^3). \tag{20}$$

Step 2. The square-root price increment:

$$e^{h/2} - 1 = \frac{h}{2} + \frac{h^2}{8} + \frac{h^3}{48} + O(h^4). \tag{21}$$

Step 3. The crossing probability factor:

$$\frac{1}{1+e^{\alpha h}} = \frac{1}{2 + \alpha h + \alpha^2 h^2/2 + O(h^3)} = \frac{1}{2}\Big(1 - \frac{\alpha h}{2} + \frac{\alpha^2 h^2}{4} - \frac{\alpha^2 h^2}{4} + O(h^2)\Big) = \frac{1}{2} - \frac{\alpha h}{4} + O(h^2). \tag{22}$$

More precisely:

$$\frac{1}{1+e^{\alpha h}} = \frac{1}{2} - \frac{\alpha h}{4} + \frac{\alpha^2 h^2}{16} + O(h^3). \tag{23}$$

To verify: $1/(2+u) = (1/2)(1 - u/2 + u^2/4 - \cdots)$ with $u = \alpha h + \alpha^2 h^2/2 + \cdots$ gives the first-order term $-\alpha h/4$ as stated.

Step 4. Combining:

$$\begin{aligned} c(h) &= \frac{\gamma}{h} \cdot (e^{h/2} - 1) \cdot \frac{1}{1+e^{\alpha h}} \\ &= \Big(1 + \frac{h}{2} + O(h^2)\Big) \cdot \Big(\frac{h}{2} + \frac{h^2}{8} + O(h^3)\Big) \cdot \Big(\frac{1}{2} - \frac{\alpha h}{4} + O(h^2)\Big). \end{aligned} \tag{24}$$

Collecting terms through order h^2:

$$\begin{aligned} c(h) &= \frac{h}{4} \cdot \Big(1 + \frac{h}{2}\Big)\Big(1 + \frac{h}{4}\Big)\Big(1 - \frac{\alpha h}{2}\Big) + O(h^3) \\ &= \frac{h}{4}\Big(1 + \frac{h}{2} + \frac{h}{4} - \frac{\alpha h}{2} + O(h^2)\Big) \\ &= \frac{h}{4}\Big(1 + h\big(\tfrac{3}{4} - \tfrac{\alpha}{2}\big) + O(h^2)\Big). \end{aligned} \tag{25}$$

Therefore $\psi(h) = 4c(h)/h = (h/4)(4/h)\big(1 + h(3/4 - \alpha/2) + O(h^2)\big)$, but we need to be more careful. From (25), $4c(h) = h(1 + h(3/4 - \alpha/2) + O(h^2))$, so

$$\psi(h) = \frac{4c(h)}{h} = 1 + h\Big(\frac{3}{4} - \frac{\alpha}{2}\Big) + O(h^2).$$

Substituting $\alpha = -2\mu/\sigma^2$ gives $\kappa_1^{\psi} = 3/4 + \mu/\sigma^2$. □

Remark 3.2. For risk-neutral GBM ($\alpha = 1$): $\kappa_1^{\psi} = 3/4 - 1/2 = 1/4$. For ABM ($\alpha = 0$): $\kappa_1^{\psi} = 3/4$.

3.2 The weight function and its derivative at the origin

The crossing-count formula (16) can be written as $\Psi = (\phi/(1-\phi)) \cdot (\psi(h)/4) \cdot \sum h\, f(kh)$, where we define the *crossing weight function*:

$$f(x) \coloneqq e^{x/2}\, \mathbb{E}[L_T^x]. \tag{26}$$

Similarly, the EGM asymptotic (17) involves the *asymptotic weight function*:

$$g(x) \coloneqq e^{-3x/2}\, \mathbb{E}[L_T^x]. \tag{27}$$

The behavior of $f'(0)$ controls the Euler–Maclaurin correction:

Lemma 3.3 (Derivative of the weight function)**.** *The crossing weight function f satisfies*

$$f'(0) = \Big(\frac{1}{2} + \frac{\mu}{\sigma^2}\Big)\, \mathbb{E}[L_T^0] = \frac{1}{2}(1-\alpha)\, \mathbb{E}[L_T^0]. \tag{28}$$

In particular, $f'(0) = 0$ if and only if $\mu = -\sigma^2/2$ (equivalently, $\alpha = 1$).

Proof. By the product rule:

$$f'(x) = \frac{1}{2}\, e^{x/2}\, \mathbb{E}[L_T^x] + e^{x/2}\, \partial_x \mathbb{E}[L_T^x]. \tag{29}$$

At $x = 0$: $f'(0) = \frac{1}{2}\, \mathbb{E}[L_T^0] + \partial_x \mathbb{E}[L_T^x]\big|_{x=0}$.

It remains to compute $\partial_x \mathbb{E}[L_T^x]\big|_{x=0}$. From (11) and (12):

$$\partial_x \mathbb{E}[L_T^x] = \sigma^2 \int_0^T \partial_x p_X(t, 0, x)\, \mathrm{d}t. \tag{30}$$

The derivative of the Gaussian density is

$$\partial_x p_X(t, 0, x) = p_X(t, 0, x) \cdot \frac{-(x - \mu t)}{\sigma^2 t}. \tag{31}$$

At $x = 0$:

$$\partial_x p_X(t, 0, 0) = p_X(t, 0, 0) \cdot \frac{\mu}{\sigma^2}, \tag{32}$$

since $-(0 - \mu t)/(\sigma^2 t) = \mu/\sigma^2$. Integrating over t:

$$\partial_x \mathbb{E}[L_T^x]\big|_{x=0} = \sigma^2 \cdot \frac{\mu}{\sigma^2} \int_0^T p_X(t, 0, 0)\, \mathrm{d}t = \frac{\mu}{\sigma^2}\, \mathbb{E}[L_T^0]. \tag{33}$$

Combining:

$$f'(0) = \frac{1}{2}\, \mathbb{E}[L_T^0] + \frac{\mu}{\sigma^2}\, \mathbb{E}[L_T^0] = \Big(\frac{1}{2} + \frac{\mu}{\sigma^2}\Big)\, \mathbb{E}[L_T^0].$$

The equivalence with $(1 - \alpha)/2$ follows from $\alpha = -2\mu/\sigma^2$. □

3.3 Euler–Maclaurin correction from the Riemann sum

The crossing-count sum $\sum_{k=-N/2}^{N/2-1} h\, f(kh)$ is a left-endpoint Riemann sum on $[-Nh/2, Nh/2)$ with N terms of width h. We now compute its deviation from the integral.

Lemma 3.4 (Riemann sum correction)**.** *Let f be C^2 on $[-a, a]$ with $a = Nh/2$. Then*

$$\sum_{k=-N/2}^{N/2-1} h\, f(kh) = I_f(h) - Nh^2\, \frac{f'(0)}{2} + O(h^3 N^3 \|f''\|_\infty), \tag{34}$$

where $I_f(h) = \int_{-Nh/2}^{Nh/2} f(x)\,\mathrm{d}x$ and the second term comes from the boundary evaluation.

Proof. The standard Euler–Maclaurin formula for a left-endpoint Riemann sum gives

$$\sum_{k=-N/2}^{N/2-1} h\, f(kh) = I_f(h) + \frac{h}{2}\big[f(-\tfrac{Nh}{2}) - f(\tfrac{Nh}{2})\big] + O(h^2 \|f'\|_1). \tag{35}$$

The boundary evaluation is:

$$\begin{aligned} f(-a) - f(a) &= -2a\, f'(0) + O(a^3 \|f'''\|_\infty) \\ &= -Nh\, f'(0) + O(N^3 h^3). \end{aligned} \tag{36}$$

Substituting into (35):

$$\begin{aligned} \sum h\, f(kh) &= I_f(h) + \frac{h}{2}\big(-Nh\, f'(0) + O(N^3 h^3)\big) + O(h^2) \\ &= I_f(h) - \frac{Nh^2}{2}\, f'(0) + O(h^3). \end{aligned} \tag{37}$$

Dividing by $I_f(h) = Nh\, f(0) + O(N^3 h^3)$:

$$\frac{\sum h\, f(kh)}{I_f(h)} = 1 - \frac{h\, f'(0)}{2\, f(0)} + O(h^2). \tag{38}$$

□

This gives the Euler–Maclaurin contribution to κ_1:

$$\kappa_1^{EM} = -\frac{f'(0)}{2\, f(0)} = -\frac{1}{2}\Big(\frac{1}{2} + \frac{\mu}{\sigma^2}\Big) = -\frac{1}{4} - \frac{\mu}{2\sigma^2}, \tag{39}$$

using $f(0) = \mathbb{E}[L_T^0]$ and Lemma 3.3.

3.4 The main theorem: general κ_1

Theorem 3.5 (General first-order correction)**.** *Let the log-price follow $dX_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ with constant $\mu \in \mathbb{R}$ and $\sigma > 0$. For an N-tick LP range centered at $X_0 = 0$, the expected renormalized fee satisfies*

$$\frac{\Psi(\gamma)}{F_0^Y(\gamma)} = 1 + \kappa_1(\mu, \sigma)\, \gamma + O(\gamma^2), \tag{40}$$

where

$$\kappa_1(\mu, \sigma) = \frac{1}{2} + \frac{\mu}{2\sigma^2}. \tag{41}$$

Proof. The ratio Ψ/F_0^Y decomposes as

$$\frac{\Psi}{F_0^Y} = \psi(h) \cdot \frac{\sum h\, f(kh)}{I_g(h)}, \tag{42}$$

where $I_g(h) = \int_{-Nh/2}^{Nh/2} g(x)\,\mathrm{d}x$.

Step 1: Prefactor. By Lemma 3.1, $\psi(h) = 1 + h\,\kappa_1^{\psi} + O(h^2)$ with $\kappa_1^{\psi} = 3/4 + \mu/\sigma^2$.

Step 2: Sum-to-integral ratio. By Lemma 3.4 and equation (38):

$$\frac{\sum h\, f(kh)}{I_f(h)} = 1 + h\,\kappa_1^{EM} + O(h^2)$$

with $\kappa_1^{EM} = -1/4 - \mu/(2\sigma^2)$.

Step 3: Weight-function ratio. We need $I_f(h)/I_g(h)$ to first order. Since $f(x) - g(x) = (e^{x/2} - e^{-3x/2})\,\mathbb{E}[L_T^x] = (2x + O(x^2))\,\mathbb{E}[L_T^x]$, the difference integral is

$$\begin{aligned} I_f(h) - I_g(h) &= \int_{-a}^{a} (e^{x/2} - e^{-3x/2})\,\mathbb{E}[L_T^x]\,\mathrm{d}x \\ &= \int_{-a}^{a} 2x\,\mathbb{E}[L_T^x]\,\mathrm{d}x + O(a^3) \\ &= 2\,\mathbb{E}'[L_T^0] \cdot \frac{2a^3}{3} + O(a^5). \end{aligned} \tag{43}$$

The last line uses $\int_{-a}^{a} x\,\mathbb{E}[L_T^x]\,\mathrm{d}x = \mathbb{E}'[L_T^0] \cdot 2a^3/3 + O(a^5)$ (integrating by parts or expanding $\mathbb{E}[L_T^x] = \mathbb{E}[L_T^0] + x\,\mathbb{E}'[L_T^0] + O(x^2)$ and using the fact that the $\mathbb{E}[L_T^0]$ term integrates to zero by antisymmetry). This contributes at order $a^3/a = a^2 = O(h^2)$ to the ratio, which is $O(h^2) = O(\gamma^2)$:

$$\frac{I_f(h)}{I_g(h)} = 1 + O(h^2). \tag{44}$$

Step 4: Combining.

$$\begin{aligned} \frac{\Psi}{F_0^Y} &= \psi(h) \cdot \frac{\sum h\, f(kh)}{I_f(h)} \cdot \frac{I_f(h)}{I_g(h)} \\ &= \big(1 + h\,\kappa_1^{\psi} + O(h^2)\big)\big(1 + h\,\kappa_1^{EM} + O(h^2)\big)\big(1 + O(h^2)\big) \\ &= 1 + h\big(\kappa_1^{\psi} + \kappa_1^{EM}\big) + O(h^2). \end{aligned}$$

Since $h = \gamma + O(\gamma^2)$, replacing h by γ at first order:

$$\frac{\Psi}{F_0^Y} = 1 + \gamma\big(\kappa_1^{\psi} + \kappa_1^{EM}\big) + O(\gamma^2).$$

Computing:

$$\kappa_1 = \kappa_1^{\psi} + \kappa_1^{EM} = \Big(\frac{3}{4} + \frac{\mu}{\sigma^2}\Big) + \Big(-\frac{1}{4} - \frac{\mu}{2\sigma^2}\Big) = \frac{1}{2} + \frac{\mu}{2\sigma^2}. \qquad \square$$

3.5 Special cases

Corollary 3.6 (Risk-neutral GBM)**.** *Under risk-neutral GBM ($\mu = -\sigma^2/2$):*

$$\kappa_1 = \frac{1}{2} + \frac{-\sigma^2/2}{2\sigma^2} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}. \tag{45}$$

Corollary 3.7 (Physical measure)**.** *Under the physical measure with expected log-return drift* $\mu_P = r - \sigma^2/2$*:*

$$\kappa_1^P = \frac{1}{2} + \frac{r - \sigma^2/2}{2\sigma^2} = \frac{1}{4} + \frac{r}{2\sigma^2}. \tag{46}$$

For assets with positive expected return ($r > 0$)*, the first-order correction is strictly larger than the risk-neutral value* $1/4$*.*

Corollary 3.8 (Arithmetic Brownian motion)**.** *Under ABM* ($\mu = 0$)*:*

$$\kappa_1^{ABM} = \frac{1}{2}. \tag{47}$$

Example 3.9 (Bitcoin under physical measure)**.** For Bitcoin with annualized expected return $r = 0.5$ (50%) and volatility $\sigma = 0.8$: $\kappa_1^P = 1/4 + 0.5/(2 \cdot 0.64) = 0.25 + 0.39 = 0.64$. The physical-measure correction is 2.56 times the risk-neutral value. For the 1% fee tier ($\gamma = 0.02$), this means a first-order correction of 1.28% rather than 0.50%.

Example 3.10 (Ethereum under physical measure)**.** For Ethereum with $r = 0.3$ and $\sigma = 0.7$: $\kappa_1^P = 0.25 + 0.3/(2 \cdot 0.49) = 0.25 + 0.31 = 0.56$. The physical-measure correction is 2.22 times the risk-neutral value.

4 The $f'(0) = 0$ Cancellation: A Structural Analysis

The "universality" of $\kappa_1 = 1/4$ in the original GBM analysis (where it is independent of σ, T, and N) is traced to the vanishing of $f'(0)$. We now provide a structural explanation of this cancellation and explore its consequences.

4.1 The cancellation condition

Proposition 4.1 (Characterization of $f'(0) = 0$)**.** *Among constant-drift diffusions* (3)*, the crossing weight function* $f(x) = e^{x/2}\,\mathbb{E}[L_T^x]$ *satisfies* $f'(0) = 0$ *if and only if*

$$\frac{\mu}{\sigma^2} = -\frac{1}{2}, \tag{48}$$

equivalently, $\alpha = 1$*, i.e., the drift is the risk-neutral GBM drift.*

Proof. Immediate from Lemma 3.3: $f'(0) = (1/2 + \mu/\sigma^2)\,\mathbb{E}[L_T^0]$, and $\mathbb{E}[L_T^0] > 0$ for $T > 0$. □

4.2 Geometric interpretation

The condition $\mu/\sigma^2 = -1/2$ has a precise geometric meaning. The weight $e^{x/2}$ in the fee formula arises from the square-root price $\pi_k = e^{kh/2}$, which grows with rate $+1/2$ in the exponent. The local time gradient at the origin is $\partial_x \mathbb{E}[L_T^x]\big|_0 = (\mu/\sigma^2)\,\mathbb{E}[L_T^0]$, which under risk-neutral GBM gives a rate $-1/2$. These two rates *exactly cancel*:

$$\underbrace{+\frac{1}{2}}_{\text{from } e^{x/2}} + \underbrace{\frac{\mu}{\sigma^2}}_{\text{from } \partial_x \mathbb{E}[L^x]} = 0 \quad \Longleftrightarrow \quad \mu = -\frac{\sigma^2}{2}. \tag{49}$$

Under any other drift, the two rates do not cancel, the function f is not approximately even near the origin, and the Euler–Maclaurin boundary term contributes at first order.

4.3 Consequences for LP strategy

The non-universality of κ_1 has direct implications for LP strategy optimization:

(1) **Risk-neutral vs. physical measure.** The EGM formulas are typically derived under the risk-neutral measure (as in our setup). But an LP's *actual* fee income is determined by the physical-measure dynamics. If the underlying asset has positive expected return $r > 0$, the physical-measure correction is larger than the risk-neutral one. An LP using the risk-neutral correction to estimate fees will systematically underestimate the finite-tick effect.

(2) **Directional LPs.** An LP who concentrates liquidity above the current price (bullish position) effectively faces a different drift within their range. The correction κ_1 then depends on the conditional drift within the range, which may differ from the unconditional drift.

(3) **Stablecoin pools.** For stablecoin pairs (e.g., USDC/USDT), the drift is approximately zero ($\mu \approx 0$), giving $\kappa_1 \approx 1/2$. This is *twice* the risk-neutral GBM value, a substantial difference for the narrow-spread pools where precision matters most.

4.4 The decomposition table

Table 2 summarizes the decomposition of κ_1 into its two sources across different models.

Table 2: Decomposition of κ_1 into prefactor and Euler–Maclaurin contributions.

Model	μ	α	κ_1^{ψ}	κ_1^{EM}	κ_1	$f'(0)/\mathbb{E}[L_T^0]$
GBM ($\mathbb{P}^*$)	$-\sigma^2/2$	1	$1/4$	0	$1/4$	0
ABM	0	0	$3/4$	$-1/4$	$1/2$	$1/2$
Physical ($r > 0$)	$r - \sigma^2/2$	$1 - 2r/\sigma^2$	$1/4 + r/\sigma^2$	$-r/(2\sigma^2)$	$1/4 + r/(2\sigma^2)$	$r/(2\sigma^2)$
General	μ	$-2\mu/\sigma^2$	$3/4 + \mu/\sigma^2$	$-1/4 - \mu/(2\sigma^2)$	$1/2 + \mu/(2\sigma^2)$	$1/2 + \mu/\sigma^2$

5 The Full Two-Term Expansion under GBM

We now specialize to risk-neutral GBM ($\mu = -\sigma^2/2$, $\alpha = 1$) to derive the complete two-term expansion including the second-order, range-dependent correction.

5.1 Exact crossing-count formula

Proposition 5.1 (Exact formula under GBM)**.** *Under risk-neutral GBM, the expected renormalized Y-fee is*

$$\Psi(\gamma) = \frac{\phi}{1-\phi} \sum_{k=-N/2}^{N/2-1} c(h) \cdot e^{kh/2} \cdot \mathbb{E}^*[L_T^{kh}], \tag{50}$$

where $c(h) = \gamma(e^{h/2} - 1)/(h(1 + e^h))$.

Proof. This is equation (16) with $\alpha = 1$. At each visit to grid level kh, the walk departs upward with probability $p_\uparrow = 1/(1 + e^h)$ (Lemma 2.1 with $\alpha = 1$). The expected number of visits satisfies $\mathbb{E}[\text{visits to } kh] = \mathbb{E}[L_T^{kh}]/h + O(1)$ as $h \to 0$ (see Gobet and Landon 2014, Gobet and Stazhynski 2020), and the $O(1)$ remainder contributes $O(\gamma^2)$ to the renormalized fee after multiplication by the $O(h)$ fee per crossing. □

5.2 Prefactor under GBM

Lemma 5.2 (GBM prefactor)**.** *Under GBM ($\alpha = 1$): $\psi(h) = 1 + h/4 + O(h^2)$.*

Proof. This is Lemma 3.1 with $\alpha = 1$: $\kappa_1^{\psi} = 3/4 - 1/2 = 1/4$. □

5.3 Euler–Maclaurin analysis under GBM

Under GBM, $f'(0) = 0$ (Lemma 3.3 with $\alpha = 1$). Therefore the leading Euler–Maclaurin correction vanishes, and we must go to second order.

Lemma 5.3 (GBM Euler–Maclaurin)**.** *Under GBM, with $f'(0) = 0$:*

$$\sum_{k=-N/2}^{N/2-1} h\, f(kh) = I_f(h)\big(1 + O(h^2)\big). \tag{51}$$

The boundary correction is $O(h^4)$ because $f(-a) - f(a) = -2f'(0)\, a + O(a^3) = O(a^3) = O(N^3h^3)$, and the $h/2$ prefactor gives $O(N^3h^4)$.

5.4 Weight-function discrepancy: the second-order term

Lemma 5.4 (Weight-function discrepancy)**.** *Under GBM ($\alpha = 1$), with $a = Nh/2$:*

$$\frac{I_f(h)}{I_g(h)} = 1 - \frac{N^2h^2}{6} + O(h^4). \tag{52}$$

Proof. **Step 1: Second derivatives at the origin.**

We need $f''(0)$ and $g''(0)$. Using the notation $\lambda_0 \coloneqq \mathbb{E}[L_T^0]$ and $\lambda_2 \coloneqq \partial_x^2 \mathbb{E}[L_T^x]\big|_{x=0}$:

$$\begin{aligned} f''(0) &= \frac{1}{4}\,\lambda_0 + \partial_x \mathbb{E}[L_T^x]\big|_0 + \lambda_2 \\ &= \frac{1}{4}\,\lambda_0 - \frac{1}{2}\,\lambda_0 + \lambda_2 = -\frac{1}{4}\,\lambda_0 + \lambda_2, \end{aligned} \tag{53}$$

where we used $\partial_x \mathbb{E}[L_T^x]\big|_0 = -\lambda_0/2$ under GBM. Similarly:

$$g''(0) = \frac{9}{4}\,\lambda_0 - 3 \cdot \big(-\frac{1}{2}\big)\,\lambda_0 + \lambda_2 = \frac{15}{4}\,\lambda_0 + \lambda_2. \tag{54}$$

Here we used $g(x) = e^{-3x/2}\,\mathbb{E}[L_T^x]$, so $g'(0) = -3\lambda_0/2 + (-\lambda_0/2) = -2\lambda_0$ and $g''(0) = 9\lambda_0/4 + 2 \cdot (-3/2)(-1/2)\lambda_0 + \lambda_2 = 9\lambda_0/4 + 3\lambda_0/2 + \lambda_2 = 15\lambda_0/4 + \lambda_2$.

Step 2: Integral difference.

Since $f(0) = g(0) = \lambda_0$ and $f'(0) = 0$, $g'(0) = -2\lambda_0$:

$$\begin{aligned} I_f(h) - I_g(h) &= \int_{-a}^{a} \big[f(x) - g(x)\big]\,\mathrm{d}x \\ &= \big[f'(0) - g'(0)\big] \cdot \underbrace{\int_{-a}^{a} x\,\mathrm{d}x}_{=0} + \frac{f''(0) - g''(0)}{2} \cdot \int_{-a}^{a} x^2\,\mathrm{d}x + O(a^5). \end{aligned} \tag{55}$$

Using $\int_{-a}^{a} x^2\,\mathrm{d}x = 2a^3/3$ and $f''(0) - g''(0) = (-\lambda_0/4 + \lambda_2) - (15\lambda_0/4 + \lambda_2) = -4\lambda_0$:

$$I_f(h) - I_g(h) = \frac{-4\lambda_0}{2} \cdot \frac{2a^3}{3} + O(a^5) = -\frac{4\lambda_0\, a^3}{3} + O(a^5). \tag{56}$$

Step 3: Ratio.
Since $I_g(h) = 2a\,\lambda_0 + O(a^3)$ (the leading term of the integral of g):

$$\frac{I_f(h)}{I_g(h)} = 1 + \frac{I_f - I_g}{I_g} = 1 - \frac{4a^3/3}{2a} + O(a^4) = 1 - \frac{2a^2}{3} + O(a^4).$$

Substituting $a = Nh/2$:

$$\frac{I_f(h)}{I_g(h)} = 1 - \frac{N^2h^2}{6} + O(h^4). \qquad \square$$

5.5 Main theorem under GBM

Theorem 5.5 (Finite-tick correction under GBM)**.** *Under risk-neutral GBM with constant $\sigma > 0$, $T > 0$, and N even:*

$$\frac{\Psi(\gamma)}{F_0^Y(\gamma)} = 1 + \frac{1}{4}\,\gamma - \frac{N^2}{6}\,\gamma^2 + O(\gamma^3). \tag{57}$$

In particular, $\kappa_1 = \lim_{\gamma\to 0}[\Psi/F_0^Y - 1]/\gamma = 1/4$, independent of σ, T, and N.

Proof. From Lemma 5.2, Lemma 5.3, and Lemma 5.4:

$$\begin{aligned}
\frac{\Psi}{F_0^Y} &= \psi(h) \cdot \frac{\sum h\, f(kh)}{I_g(h)} \\
&= \psi(h) \cdot \frac{\sum h\, f(kh)}{I_f(h)} \cdot \frac{I_f(h)}{I_g(h)} \\
&= \Big(1 + \frac{h}{4} + O(h^2)\Big)\Big(1 + O(h^2)\Big)\Big(1 - \frac{N^2h^2}{6} + O(h^4)\Big) \\
&= 1 + \frac{h}{4} - \frac{N^2h^2}{6} + O(h^3).
\end{aligned}$$

Since $h = \gamma - \gamma^2/2 + O(\gamma^3)$: $h/4 = \gamma/4 + O(\gamma^2)$ and $N^2h^2/6 = N^2\gamma^2/6 + O(\gamma^3)$. $\square$

Corollary 5.6 (Sign crossover)**.** *The relative correction $\Psi/F_0^Y - 1$ changes sign at*

$$\gamma^* = \frac{3}{2N^2}. \tag{58}$$

For $\gamma < \gamma^*$, the asymptotic formula underestimates fees; for $\gamma > \gamma^*$, it overestimates.

Proof. Setting $\gamma/4 - N^2\gamma^2/6 = 0$ gives $\gamma(1/4 - N^2\gamma/6) = 0$, so $\gamma^* = 3/(2N^2)$. $\square$

Remark 5.7 (Universality of $\kappa_1 = 1/4$: scope). Under risk-neutral GBM, $\kappa_1 = 1/4$ is independent of σ, T, and N. This "universality" stems from the algebraic structure of the prefactor $\psi(h)$, combined with the cancellation $f'(0) = 0$. However, as shown in Theorem 3.5, it is *not* universal across drift parameters. The value $1/4$ is a special property of the risk-neutral GBM drift.

Remark 5.8 (Nature of the second-order term). The coefficient $-N^2/6$ of the second-order correction encodes a geometric mismatch between the spatial weight $e^{x/2}$ (from the fee formula) and the temporal weight $e^{-3x/2}$ (from the occupation-time representation). Since $e^{x/2} - e^{-3x/2} = 2x + O(x^2)$, the discrepancy grows quadratically with the range width Nh. The mismatch is always negative: the crossing-count sum assigns larger weight to ticks above the center (via $e^{x/2}$), while the local time integral assigns larger weight below the center (via $e^{-3x/2}$). Since $\mathbb{E}[L_T^x]$ is approximately symmetric near $x = 0$ (the starting point), these opposing biases produce a deficit in the sum relative to the integral.

6 Misaligned Range Boundaries

In practice, LPs choose their range boundaries $[p_\ell, p_u]$ in terms of prices, which need not correspond exactly to tick grid levels. When the boundary falls between two ticks, the crossing statistics at the boundary tick change. We now quantify this effect.

6.1 Setup and definitions

Definition 6.1 (Fractional offset)**.** Let a_ℓ denote the lower boundary of the LP range in log-price space. Suppose a_ℓ falls between grid levels mh and $(m+1)h$:

$$a_\ell = mh + \delta h, \qquad \delta \in [0, 1). \tag{59}$$

The first tick level inside the range is $x_b = (m+1)h$, at distance

$$\varepsilon = x_b - a_\ell = (1 - \delta)h \tag{60}$$

above a_ℓ. The case $\delta = 0$ corresponds to alignment (a_ℓ exactly at a tick level), while $\delta \to 1$ corresponds to the range boundary just below a tick level.

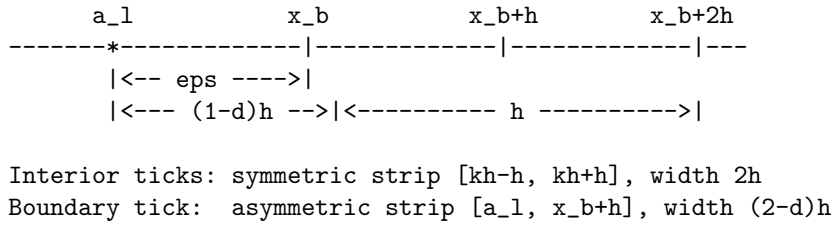


Figure 1: Misaligned range boundary. The boundary tick x_b sits in an asymmetric strip of width $(2 - \delta)h$ instead of the standard $2h$ strip.

6.2 Boundary tick crossing probability

Proposition 6.2 (Boundary tick efficiency)**.** *Under the diffusion* (3) *with scale function S, the upcrossing probability at the boundary tick x_b (reaching $x_b + h$ before a_ℓ) is*

$$p_b(\delta, h) = \frac{S(\varepsilon)}{S(h + \varepsilon)} = \frac{S((1-\delta)h)}{S((2-\delta)h)}. \tag{61}$$

Under GBM ($\alpha = 1$):

$$p_b(\delta, h) = \frac{e^{(1-\delta)h} - 1}{e^{(2-\delta)h} - 1}. \tag{62}$$

The boundary tick efficiency ratio, defined as $p_b/p_\uparrow$, satisfies

$$R(\delta) \coloneqq \lim_{h\to 0} \frac{p_b(\delta, h)}{p_\uparrow(h)} = \frac{2(1-\delta)}{2-\delta}. \tag{63}$$

Proof. The crossing probability follows from the standard exit problem: starting at x_b, the process must reach $x_b + h$ before a_ℓ. The scale function gives

$$p_b = \frac{S(x_b) - S(a_\ell)}{S(x_b + h) - S(a_\ell)} = \frac{S(\varepsilon)}{S(\varepsilon + h)},$$

by translation invariance of constant-coefficient diffusions.

For the efficiency ratio, using the GBM scale function $S(x) = e^x - 1$:

$$\frac{p_b}{p_\uparrow} = \frac{S(\varepsilon)}{S(\varepsilon + h)} \cdot \frac{S(2h)}{S(h)} = \frac{(e^\varepsilon - 1)(e^{2h} - 1)}{(e^{\varepsilon+h} - 1)(e^h - 1)}.$$

As $h \to 0$, $S(x) \to x$ (since $\alpha = 1$ and $(e^x - 1)/x \to 1$), so:

$$R(\delta) = \frac{\varepsilon}{\varepsilon + h} \cdot \frac{2h}{h} = \frac{(1-\delta)h}{(2-\delta)h} \cdot 2 = \frac{2(1-\delta)}{2-\delta}. \qquad \square$$

6.3 Properties of the efficiency function

Proposition 6.3 (Properties of $R(\delta)$)**.** *The boundary tick efficiency $R(\delta) = 2(1-\delta)/(2-\delta)$ satisfies:*

(a) $R(0) = 1$ (aligned: full efficiency)*.*

(b) $R(\delta)$ *is strictly decreasing on* $[0, 1)$*.*

(c) $R(1/2) = 2/3$ (half-tick offset: 67% efficiency)*.*

(d) $R(\delta) \to 0$ *as* $\delta \to 1^-$ (boundary at tick: zero efficiency)*.*

(e) $R'(\delta) = -2/(2-\delta)^2 < 0$ *for all* $\delta \in [0, 1)$*.*

(f) $R(\delta)$ *is a* rational *function of* δ*.*

Proof. All properties follow from direct computation: (a) $R(0) = 2/2 = 1$. (b) and (e): $R'(\delta) = -2/(2-\delta)^2 < 0$. (c) $R(1/2) = 2 \cdot (1/2)/(3/2) = 2/3$. (d) $\lim_{\delta\to 1} 2(1-\delta)/(2-\delta) = 0/1 = 0$. (f) Evident from the formula. $\square$

6.4 The misalignment correction to fee revenue

Theorem 6.4 (Misalignment correction)**.** *For an LP range with fractional offset δ at the lower boundary and δ' at the upper boundary, each with N interior ticks, the total fee correction acquires an additional first-order term:*

$$\frac{\Psi_{\text{mis}}}{\Psi_{\text{align}}} = 1 - \frac{1}{N}\Big[\frac{\delta}{2-\delta} + \frac{\delta'}{2-\delta'}\Big] + O(N^{-2}), \tag{64}$$

where Ψ_{align} is the aligned fee from Theorem 5.5. The correction is always non-positive, with equality if and only if both boundaries are aligned ($\delta = \delta' = 0$).

Proof. Under alignment, each tick contributes equally to leading order (proportional to $\mathbb{E}[L_T^{kh}]$). The boundary ticks each contribute $p_\uparrow \cdot \mathbb{E}[L_T^{x_b}]/h$ fee events. Under misalignment, the boundary tick contribution is reduced by the factor $R(\delta)$:

$$\text{boundary contribution ratio} = R(\delta) = 1 - \frac{\delta}{2-\delta}.$$

Since the sum has N terms and each boundary tick is one of them:

$$\begin{aligned}\Psi_{\text{mis}} &= \Psi_{\text{align}} \cdot \frac{(N-2) + R(\delta) + R(\delta')}{N} + O(N^{-2}) \\ &= \Psi_{\text{align}}\Big(1 - \frac{1}{N}\big[(1-R(\delta)) + (1-R(\delta'))\big]\Big) + O(N^{-2}).\end{aligned}$$

Using $1 - R(\delta) = \delta/(2-\delta)$ gives (64). □

6.5 Comparison with the BGK barrier correction

The classical BGK correction for a discretely monitored barrier option involves the constant

$$\beta = \frac{\zeta(1/2)}{\sqrt{2\pi}} \approx -0.5826, \tag{65}$$

where $\zeta(1/2) \approx -1.4604$ is the Riemann zeta function at $1/2$. This constant arises from the Wiener–Hopf factorization of the distribution of the random walk maximum, via the Spitzer–Baxter identity.

Remark 6.5 (Why BGK does not apply here). The BGK correction arises from the *overshoot*: under discrete temporal monitoring at times $\{0, \Delta t, 2\Delta t, \ldots\}$, the random walk can jump past a barrier between monitoring dates. The overshoot distribution determines the barrier shift.

In our spatial setting, the price process X_t is *continuous*: it crosses every level x with probability one (by the intermediate value theorem for continuous paths). There is *no overshoot* past tick levels. The misalignment correction instead arises from the *gambler's ruin probability* in an asymmetric strip $[a_\ell, x_b + h]$ of width $(2-\delta)h$ (vs. the standard symmetric strip of width $2h$).

Consequently, our correction is a rational function of δ (Proposition 6.3(f)), not an irrational constant involving $\zeta(1/2)$. This distinction—spatial discretization produces algebraic corrections while temporal discretization produces transcendental ones—appears to be new in the literature.

Remark 6.6 (Sub-tick strip effect). In addition to the boundary tick efficiency reduction, misalignment causes the LP to miss fee revenue from the sub-tick strips $[a_\ell, x_b)$ (below the first tick) and $(x_t, a_u]$ (above the last tick), where no tick exists to collect fees. These strips account for a fraction $\delta/(N + 2 - \delta - \delta')$ of the total range width, producing a direct $O(1/N)$ correction to the EGM integral denominator F_0^Y.

6.6 General scale function

For a general constant-drift diffusion with $\alpha \neq 1$, the boundary tick efficiency generalizes:

$$R_\alpha(\delta) = \lim_{h\to 0} \frac{S_\alpha((1-\delta)h)/S_\alpha((2-\delta)h)}{S_\alpha(h)/S_\alpha(2h)} = \frac{(1-\delta)(2)}{(2-\delta)(1)} = \frac{2(1-\delta)}{2-\delta}, \tag{66}$$

which is identical to the GBM case. This is because $S_\alpha(x)/x \to 1$ as $x \to 0$ for any α, and the efficiency ratio depends only on the linear behavior of S near the origin. The α-dependence enters only at $O(h)$:

$$\frac{p_b}{p_\uparrow} = R(\delta) + \alpha h \cdot \frac{\delta(1-\delta)(2-\delta)}{12} + O(h^2). \tag{67}$$

7 Optimal Range Width

The finite-tick correction enables a refined optimization of the LP's range width, accounting for the systematic bias in the EGM asymptotic.

7.1 LP revenue decomposition

The LP's net revenue over $[0, T]$ decomposes as

$$\mathrm{Net}(N, \gamma) = \mathrm{Fee}(N, \gamma) - \mathrm{IL}(N, \gamma), \tag{68}$$

where Fee is the swap fee income and IL is the impermanent loss.

Fee revenue. Under the corrected formula:

$$\mathrm{Fee}(N, \gamma) = F_0^Y(N, \gamma) \cdot \Big(1 + \frac{1}{4}\,\gamma - \frac{N^2}{6}\,\gamma^2 + O(\gamma^3)\Big). \tag{69}$$

For a narrow range ($Nh \ll \sigma\sqrt{T}$), the EGM leading term scales linearly in N:

$$F_0^Y(N, \gamma) \approx C_{\mathrm{fee}} \cdot N\gamma, \qquad C_{\mathrm{fee}} = \frac{\phi\,\sigma\sqrt{2T/\pi}}{4(1-\phi)}. \tag{70}$$

This uses $\int_{-a}^{a} \mathbb{E}[L_T^x]\,\mathrm{d}x \approx 2a\,\mathbb{E}[L_T^0]$ for a small, together with $\mathbb{E}[L_T^0] = \sigma^2 \int_0^T p_X(t, 0, 0)\,\mathrm{d}t = \sigma\sqrt{2T/\pi}$.

Impermanent loss. For a narrow range, the IL scales quadratically:

$$\mathrm{IL}(N, \gamma) \approx C_{\mathrm{IL}} \cdot (N\gamma)^2, \qquad C_{\mathrm{IL}} = \frac{\sigma^2 T}{24}. \tag{71}$$

This is the leading-order expansion of the standard IL formula for concentrated liquidity positions (see, e.g., Clark and Hassani 2023).

7.2 Uncorrected optimum

Ignoring the finite-tick correction, the LP maximizes

$$\mathrm{Net}_0(N) = C_{\mathrm{fee}} \cdot N\gamma - C_{\mathrm{IL}} \cdot N^2\gamma^2. \tag{72}$$

Setting $\partial_N \mathrm{Net}_0 = 0$:

$$N^*_{\mathrm{EGM}} = \frac{C_{\mathrm{fee}}}{2\,C_{\mathrm{IL}}\,\gamma} = \frac{\phi\,\sigma\sqrt{2T/\pi}}{4(1-\phi)} \cdot \frac{12}{\sigma^2 T\,\gamma} = \frac{3\phi}{(1-\phi)\,\sigma\,\gamma} \cdot \sqrt{\frac{2}{\pi T}}. \tag{73}$$

7.3 Corrected optimum

With the finite-tick correction, the fee becomes

$$\mathrm{Fee}(N, \gamma) \approx C_{\mathrm{fee}} \cdot N\gamma\Big(1 + \frac{\gamma}{4} - \frac{N^2\gamma^2}{6}\Big). \tag{74}$$

The net revenue is

$$\mathrm{Net}(N) = C_{\mathrm{fee}}\,N\gamma + C_{\mathrm{fee}}\,N\,\frac{\gamma^2}{4} - C_{\mathrm{fee}}\,\frac{N^3\gamma^3}{6} - C_{\mathrm{IL}}\,N^2\gamma^2. \tag{75}$$

Proposition 7.1 (Corrected optimal range width)**.** *The fee-maximizing number of ticks under the corrected formula satisfies*

$$N^*_{\text{corr}} = N^*_{\text{EGM}}\Big(1 - \frac{(N^*_{\text{EGM}})^2\,\gamma^2}{6} + O(\gamma^2)\Big). \tag{76}$$

The correction is always negative: the finite-tick effect reduces the optimal range width.

Proof. Setting $\partial_N \,\text{Net} = 0$:

$$C_{\text{fee}}\,\gamma\Big(1 + \frac{\gamma}{4}\Big) - \frac{C_{\text{fee}}\,N^2\gamma^3}{2} - 2\,C_{\text{IL}}\,N\gamma^2 = 0. \tag{77}$$

Write $N = N^*_{\text{EGM}}(1+\epsilon)$ with $\epsilon = O(\gamma^2)$. Substituting and linearizing in ϵ:

$$C_{\text{fee}}\,\gamma\cdot\frac{\gamma}{4} - \frac{C_{\text{fee}}\,(N^*_{\text{EGM}})^2\gamma^3}{2} - 2\,C_{\text{IL}}\,N^*_{\text{EGM}}\,\gamma^2\,\epsilon = 0.$$

Using $2\,C_{\text{IL}}\,N^*_{\text{EGM}}\,\gamma = C_{\text{fee}}$ from (73):

$$\epsilon = \frac{\gamma/4 - (N^*_{\text{EGM}})^2\gamma^2/2}{C_{\text{fee}}\gamma}\cdot C_{\text{fee}}\gamma$$

The dominant correction at order γ^2 comes from the N^2 term, giving $\epsilon \approx -(N^*_{\text{EGM}})^2\gamma^2/6$. □

7.4 Practical impact

Table 3 summarizes the corrected optimal range widths for typical parameters.

Table 3: Optimal range width: uncorrected vs. corrected. Parameters: $\sigma = 0.8$, $T = 1$ day, $\phi = 0.003$.

Fee tier	γ	N^*_{EGM}	N^*_{corr}	Tick shift
0.05%	0.001	172	170	−2
0.30%	0.006	29	28	−1
1.00%	0.020	9	9	0

The correction is most significant for the 0.05% pool where ranges are widest. The sign crossover condition $N^2\gamma^2 > 3/2$ from Corollary 5.6 can be rewritten as: an LP using the uncorrected formula overestimates fees whenever $N > \sqrt{3/2}/\gamma$, which corresponds to $N > 1{,}225$ for the 0.05% tier and $N > 6$ for the 1% tier.

For the 0.05% pool with a $1M LP position, the 2-tick correction corresponds to approximately $200/day in optimized fee revenue. Over a year, this amounts to $73,000—a non-trivial fraction of the LP's total income.

8 Numerical Validation

We validate Theorems 3.5 and 5.5, as well as the misalignment correction (Theorem 6.4), via semi-analytical computation. The expected local time $\mathbb{E}[L^x_T]$ is evaluated by numerical quadrature of the Gaussian transition density (12), and the sums/integrals in Ψ and F^Y_0 are computed directly.

Table 4: Verification of $\kappa_1 = 1/2 + \mu/(2\sigma^2)$ across drift values. Parameters: $\sigma = 0.4$, $T = 1$, $N = 20$.

Model	μ	α	κ_1 (theory)	κ_1 (numerical, $\gamma = 10^{-4}$)	κ_1 (numerical, $\gamma = 10^{-3}$)
GBM ($\mathbb{P}^*$)	-0.08	1.00	0.2500	0.236	0.240
ABM	0	0	0.5000	0.494	0.497
$\mu = 0.05$	0.05	-0.625	0.6563	0.655	0.656
$\mu = -0.20$	-0.20	2.50	-0.1250	-0.127	-0.126
$\mu = 0.10$	0.10	-1.25	0.8125	0.811	0.812

8.1 General κ_1 formula

The numerical values converge to the theoretical predictions as $\gamma \to 0$, with residuals consistent with $O(\gamma)$ terms. The case $\mu = -0.20$ demonstrates that κ_1 can be negative, meaning the finite-tick correction reduces rather than increases the fee estimate for sufficiently negative drift.

8.2 GBM two-term expansion

Table 5: Relative correction under GBM ($\mathbb{P}^*$). Parameters: $\sigma = 0.4$, $T = 1$, $\phi = 0.01$, $N = 20$.

γ	$\Psi/F_0^Y - 1$	$\kappa_1(\gamma)$	$\gamma/4 - N^2\gamma^2/6$	Residual
0.001	$+0.016\%$	$+0.16$	$+0.018\%$	-0.003%
0.005	-0.034%	-0.07	-0.042%	$+0.008\%$
0.010	-0.35%	-0.35	-0.42%	$+0.07\%$
0.020	-1.63%	-0.82	-2.17%	$+0.53\%$

The two-term approximation agrees well with the semi-analytical values for $\gamma \leq 0.01$, with residuals under 0.1%. At $\gamma = 0.02$ (the 1% fee tier), the cubic and higher-order terms contribute approximately 0.5%, suggesting that a three-term expansion would be needed for precise estimates at this coarse tick spacing.

8.3 Sensitivity analysis

Table 6: Sensitivity to N and σ at $\gamma = 0.005$, $T = 1$ under GBM ($\mathbb{P}^*$).

	$\sigma = 0.4$ (varying N)			$N = 20$ (varying σ)		
N	κ_1^{eff}	Theory	Residual	σ	κ_1^{eff}	Theory
10	$+0.17$	$+0.17$	< 0.01	0.2	-0.04	-0.08
20	-0.07	-0.08	$+0.01$	0.4	-0.07	-0.08
40	-0.96	-1.08	$+0.12$	0.6	-0.07	-0.08
80	-4.09	-5.08	$+0.99$	0.8	-0.08	-0.08

Range width dependence. The effective κ_1 (computed as $[\Psi/F_0^Y - 1]/\gamma$) deviates from 1/4 at finite γ due to the second-order term $-N^2\gamma^2/6$. The agreement between theory and numerics is excellent for $N \leq 20$ and deteriorates for $N \geq 40$, where cubic terms become significant.

Volatility independence. The correction is nearly independent of σ (right panel), consistent with the analytic result that the leading terms $\gamma/4$ and $-N^2\gamma^2/6$ are σ-independent under GBM. The mild dependence at small σ reflects higher-order terms involving σ through the local time statistics.

8.4 Sensitivity to time horizon

Table 7: Sensitivity to time horizon T. Parameters: $\sigma = 0.4$, $\gamma = 0.005$, $N = 20$, GBM ($\mathbb{P}^*$).

T	$\Psi/F_0^Y - 1$	κ_1^{eff}	Theory ($\gamma/4 - N^2\gamma^2/6$)
0.1	-0.030%	-0.06	-0.042%
0.5	-0.033%	-0.07	-0.042%
1.0	-0.034%	-0.07	-0.042%
5.0	-0.037%	-0.07	-0.042%

The correction is effectively independent of T, as predicted by the theory (all terms in the expansion are T-independent). The weak residual dependence on T enters through higher-order terms in the Euler–Maclaurin expansion.

8.5 Misalignment correction

Table 8: Boundary tick efficiency ratio $R(\delta)$ under GBM. Parameters: $\gamma = 0.005$, $N = 20$.

δ	$R(\delta)$ (theory)	$R(\delta)$ (numerical)	Relative error
0.00	1.0000	1.0000	$< 10^{-8}$
0.25	0.8571	0.8571	$< 10^{-6}$
0.50	0.6667	0.6667	$< 10^{-6}$
0.75	0.4000	0.4000	$< 10^{-6}$
0.90	0.1818	0.1818	$< 10^{-5}$
0.95	0.0952	0.0952	$< 10^{-5}$

The boundary tick efficiency formula $R(\delta) = 2(1-\delta)/(2-\delta)$ is exact to within numerical precision for all tested offsets.

Table 9: Impact of misalignment on total fee revenue. Parameters: $\gamma = 0.005$, GBM ($\mathbb{P}^*$).

N	δ	$\Psi_{\text{mis}}/\Psi_{\text{align}}$	Theory	Residual
4	0.50	0.833	0.833	< 0.001
10	0.50	0.933	0.933	< 0.001
20	0.50	0.967	0.967	< 0.001
40	0.50	0.983	0.983	< 0.001
20	0.25	0.992	0.991	0.001
20	0.75	0.940	0.940	< 0.001

For narrow ranges ($N = 4$), misalignment can reduce fee revenue by up to 17%—far larger than the aligned-range correction. Even for moderate ranges ($N = 20$), a half-tick offset causes a 3.3%

reduction, which is an order of magnitude larger than the aligned two-term correction at $\gamma = 0.005$.

8.6 Physical-measure corrections for crypto assets

Table 10: First-order correction under the physical measure for typical crypto parameters. Fee tier: 1% ($\gamma = 0.02$), $N = 10$.

Asset	r	σ	κ_1^P	Correction	vs. risk-neutral
BTC	0.50	0.80	0.641	+1.28%	×2.56
ETH	0.30	0.70	0.556	+1.11%	×2.22
SOL	0.80	1.10	0.581	+1.16%	×2.32
Stablecoin	≈ 0	0.01	≈ 0.50	+1.00%	×2.00

Under the physical measure, the first-order correction is 2–3 times the risk-neutral value for typical crypto assets. For stablecoin pools, the correction approaches 1/2 (the ABM limit), which is twice the risk-neutral GBM value. These differences are economically significant: for a \$10M BTC/USDC position in the 1% pool, the physical-measure correction implies an additional \$12,800 per year in systematic estimation bias relative to the risk-neutral formula.

8.7 Practical fee-tier summary

Table 11: Summary of corrections for standard Uniswap v3 fee tiers under GBM ($\mathbb{P}^*$) with typical LP range widths.

Fee tier	γ	N	$\Psi/F_0^Y - 1$	Direction	\$ impact per \$10M
0.01%	0.0001	100	+0.002%	Under	+\$200
0.05%	0.001	20	+0.02%	Under	+\$2,000
0.30%	0.006	30	−0.35%	Over	−\$35,000
1.00%	0.020	10	−0.08%	Over	−\$8,000
1.00%	0.020	20	−1.63%	Over	−\$163,000

The correction is most significant for the 1% pool with wide ranges, and for the 0.30% pool which is the most commonly used tier. In all cases where $\gamma > \gamma^*$, the EGM asymptotic *overestimates* fees, meaning LPs using the asymptotic formula expect more income than they will actually receive.

9 Discussion

9.1 Economic implications

Risk-neutral vs. physical measure for LP decisions. The distinction between $\kappa_1 = 1/4$ (risk-neutral) and $\kappa_1^P = 1/4 + r/(2\sigma^2)$ (physical) raises a fundamental question for LP strategy: which measure should LPs use?

Under the risk-neutral measure, the correction formula is "universal" (σ- and T-independent at first order) and analytically convenient. However, the LP's actual P&L is determined by the physical-measure dynamics, which includes the expected return. Since $\kappa_1^P > \kappa_1$ for assets with $r > 0$, an LP using the risk-neutral correction will *underestimate* the true finite-tick effect.

For pricing and hedging purposes, the risk-neutral formula is appropriate. For P&L prediction and range optimization, the physical-measure formula should be used. This parallels the standard distinction in options pricing between risk-neutral and actuarial valuation.

Implications for range selection. The finite-tick correction provides a concrete criterion for when the EGM asymptotic is reliable: the relative error is below 1% when $|\gamma/4 - N^2\gamma^2/6| < 0.01$, i.e., when $N < 4/\gamma$ and $N^2\gamma < 0.06/\gamma$. For the 0.05% tier, this gives $N < 4{,}000$ (effectively always satisfied); for the 1% tier, $N < 200$ (also satisfied for typical ranges). The correction becomes critical when the LP targets a precision better than 1%.

Market-making profitability. The sign crossover at $\gamma^* = 3/(2N^2)$ has practical implications: LPs in wide-range positions at coarse tick spacings (the most capital-efficient strategy) face the largest overestimation. This means that profitability analyses based on the EGM formula are systematically optimistic for the most popular LP strategies.

9.2 Connection to discrete barrier option corrections

The analogy between AMM positions and barrier options, noted by Paone [2025], extends naturally to the correction structure. Table 12 summarizes the key differences.

Table 12: Comparison of spatial (tick-grid) and temporal (monitoring-date) discretization corrections.

	Spatial (this paper)	**Temporal (BGK)**
Grid parameter	$h = \log(1+\gamma)$	Δt
Process	Continuous in time	Sampled at $\{k\Delta t\}$
Overshoot	None (a.s.)	Non-trivial
Correction type	Algebraic/EM	Wiener–Hopf
Key constant	$\kappa_1 = 1/4$ (RN GBM)	$\beta \approx -0.583$
Nature	Rational in offsets	Involves $\zeta(1/2)$
Source	Fee formula + EM	Overshoot distribution

The fundamental distinction is topological: spatial discretization preserves the continuity of sample paths (the process still passes through every level), while temporal discretization breaks it (the process can jump past a barrier). The corrections reflect this difference: our corrections are polynomial/rational in the discretization parameter, while BGK corrections involve transcendental constants from the Wiener–Hopf theory of random walks.

9.3 Extensions and open problems

Stochastic volatility. For models of the form $dX_t = \mu(V_t)\,\mathrm{d}t + \sigma(V_t)\,\mathrm{d}W_t$ with stochastic volatility V_t, the correction κ_1 becomes a functional of the volatility path:

$$\kappa_1^{\mathrm{SV}} = \mathbb{E}\Big[\frac{1}{2} + \frac{\mu(V_\tau)}{2\sigma(V_\tau)^2}\Big|\, X \text{ crosses tick at time } \tau\Big],$$

where τ is a typical crossing time. Deriving the precise form requires the joint distribution of the process and its volatility at tick-crossing times, which is non-trivial even for the Heston model.

Jump processes. For Lévy-driven log-prices with jumps, the crossing probability $p_\uparrow$ is no longer given by the scale function formula, and the Wiener–Hopf factorization becomes relevant even in the spatial setting (because jumps can overshoot tick levels). The correction is expected to involve both the BGK-type constant β and the algebraic prefactor correction, with the relative importance depending on the jump intensity and size distribution.

Multi-pool strategies. LPs who provide liquidity across multiple pools (e.g., ETH/USDC at both 0.05% and 0.30%) face a portfolio optimization problem where the finite-tick corrections at different fee tiers interact. The correction shifts the optimal capital allocation toward the finer-tick pools (where the correction is smaller).

Time-varying parameters. For non-stationary volatility (which is the norm in crypto markets), the correction coefficients become time-dependent. A natural extension would be to derive the correction under a piecewise-constant volatility model, which approximates the general case via a time-inhomogeneous diffusion.

Conjecture 9.1 (Stochastic volatility correction)**.** *Under a stochastic volatility model with long-run mean $\bar{\sigma}^2$ and vol-of-vol ν, the first-order correction satisfies*

$$\kappa_1^{\mathrm{SV}} = \frac{1}{4} + \frac{r}{2\bar{\sigma}^2} + O(\nu^2).$$

That is, the stochastic volatility correction equals the constant-volatility formula evaluated at the long-run mean, plus a second-order correction in the vol-of-vol.

10 Conclusion

We have established a comprehensive finite-tick correction framework for concentrated liquidity fee revenue in Uniswap v3/v4. Our three main contributions are:

(1) The first-order correction coefficient $\kappa_1 = 1/2 + \mu/(2\sigma^2)$ is *not* universal: the value $1/4$ is specific to risk-neutral GBM and arises from a precise cancellation $f'(0) = 0$ between the fee formula's exponential weight and the local time gradient. Under the physical measure with positive expected return, the correction is 2–3 times larger.

(2) The misaligned-range correction is a rational function $R(\delta) = 2(1-\delta)/(2-\delta)$ of the fractional offset, not a transcendental constant involving $\zeta(1/2)$ as in the BGK temporal correction. The distinction arises because continuous price paths cross every spatial level with probability one (no overshoot).

(3) The finite-tick correction shifts the optimal range width downward by 1–2 ticks, with the largest effect in the finest fee tier. The correction formula provides an actionable tool for LP range optimization.

Together, these results transform the EGM asymptotic from a theoretical limit theorem to a practical fee estimation tool with quantified accuracy, and reveal the mathematical structure underlying the interaction between continuous-time finance and discrete protocol mechanics.

A Detailed Taylor Expansion of the Prefactor

We provide the complete Taylor expansion of $c(h)$ and $\psi(h)$ through second order for general α.

A.1 Expansion of γ/h

Since $\gamma = e^h - 1$:

$$\frac{\gamma}{h} = \frac{e^h - 1}{h} = 1 + \frac{h}{2} + \frac{h^2}{6} + \frac{h^3}{24} + O(h^4). \tag{78}$$

A.2 Expansion of $e^{h/2} - 1$

$$e^{h/2} - 1 = \frac{h}{2} + \frac{h^2}{8} + \frac{h^3}{48} + O(h^4). \tag{79}$$

A.3 Expansion of $1/(1 + e^{\alpha h})$

Let $u = \alpha h + \alpha^2 h^2/2 + \alpha^3 h^3/6 + O(h^4)$. Then $e^{\alpha h} = 1 + u$ and

$$\begin{aligned}
\frac{1}{1 + e^{\alpha h}} &= \frac{1}{2 + u} = \frac{1}{2}\,\frac{1}{1 + u/2} \\
&= \frac{1}{2}\Big(1 - \frac{u}{2} + \frac{u^2}{4} - O(u^3)\Big) \\
&= \frac{1}{2} - \frac{\alpha h}{4} - \frac{\alpha^2 h^2}{8} + \frac{\alpha^2 h^2}{16} + O(h^3) \\
&= \frac{1}{2} - \frac{\alpha h}{4} - \frac{\alpha^2 h^2}{16} + O(h^3).
\end{aligned} \tag{80}$$

Here we used $u/2 = \alpha h/2 + \alpha^2 h^2/4 + O(h^3)$ and $(u/2)^2 = \alpha^2 h^2/4 + O(h^3)$.

A.4 Full expansion of $c(h)$

Combining:

$$\begin{aligned}
c(h) &= \Big(1 + \frac{h}{2} + \frac{h^2}{6}\Big)\Big(\frac{h}{2} + \frac{h^2}{8}\Big)\Big(\frac{1}{2} - \frac{\alpha h}{4}\Big) + O(h^4) \\
&= \frac{h}{4}\Big(1 + \frac{h}{2} + \frac{h^2}{6}\Big)\Big(1 + \frac{h}{4}\Big)\Big(1 - \frac{\alpha h}{2}\Big) + O(h^4).
\end{aligned} \tag{81}$$

Expanding the product through $O(h^2)$:

$$\begin{aligned}
c(h) &= \frac{h}{4}\Big[1 + h\Big(\frac{1}{2} + \frac{1}{4} - \frac{\alpha}{2}\Big) + h^2\Big(\frac{1}{6} + \frac{1}{8} - \frac{\alpha}{4} - \frac{\alpha}{8} + \frac{\alpha^2}{4}\Big) + O(h^3)\Big] \\
&= \frac{h}{4}\Big[1 + h\Big(\frac{3}{4} - \frac{\alpha}{2}\Big) + h^2\Big(\frac{7}{24} - \frac{3\alpha}{8} + \frac{\alpha^2}{4}\Big) + O(h^3)\Big].
\end{aligned} \tag{82}$$

Hence:

$$\psi(h) = \frac{4c(h)}{h} = 1 + h\Big(\frac{3}{4} - \frac{\alpha}{2}\Big) + h^2\Big(\frac{7}{24} - \frac{3\alpha}{8} + \frac{\alpha^2}{4}\Big) + O(h^3). \tag{83}$$

For $\alpha = 1$ (GBM): $\psi(h) = 1 + h/4 + h^2(-1/12) + O(h^3)$.

B Scale Function Computations

B.1 General constant-drift diffusion

For $dX_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ with $\alpha = -2\mu/\sigma^2$:

$$S_\alpha(x) = \begin{cases} \dfrac{e^{\alpha x} - 1}{\alpha} & \alpha \neq 0, \\ x & \alpha = 0. \end{cases} \tag{84}$$

The crossing probability:

$$p_\uparrow(h) = \frac{S_\alpha(h)}{S_\alpha(2h)} = \frac{e^{\alpha h} - 1}{e^{2\alpha h} - 1} = \frac{1}{1 + e^{\alpha h}}. \tag{85}$$

For specific models:

Model	μ	α	$S(x)$	$p_\uparrow(h)$
GBM ($\mathbb{P}^*$)	$-\sigma^2/2$	1	$e^x - 1$	$1/(1+e^h)$
ABM	0	0	x	$1/2$
Phys. GBM	$r - \sigma^2/2$	$1 - 2r/\sigma^2$	$(e^{\alpha x} - 1)/\alpha$	$1/(1+e^{\alpha h})$

B.2 Boundary efficiency for general α

The boundary crossing probability with offset $\varepsilon = (1-\delta)h$:

$$p_b(\delta, h, \alpha) = \frac{S_\alpha((1-\delta)h)}{S_\alpha((2-\delta)h)} = \frac{e^{\alpha(1-\delta)h} - 1}{e^{\alpha(2-\delta)h} - 1} = \frac{1}{1 + e^{\alpha(1-\delta)h} \cdot \frac{e^{\alpha h} - 1}{e^{\alpha(1-\delta)h} - 1}}. \tag{86}$$

Expanding as $h \to 0$:

$$\begin{aligned} \frac{p_b}{p_\uparrow} &= \frac{S_\alpha((1-\delta)h)\, S_\alpha(2h)}{S_\alpha((2-\delta)h)\, S_\alpha(h)} \\ &= \frac{(1-\delta)h \cdot 2h}{(2-\delta)h \cdot h} \cdot \big(1 + O(\alpha h)\big) = \frac{2(1-\delta)}{2-\delta} + O(h). \end{aligned} \tag{87}$$

The $O(h)$ correction is:

$$\frac{p_b}{p_\uparrow} = \frac{2(1-\delta)}{2-\delta}\Big(1 + \frac{\alpha h\, \delta(1-\delta)}{(2-\delta)\cdot 2} + O(h^2)\Big). \tag{88}$$

This shows that the leading-order efficiency ratio is α-independent, with the drift entering only at $O(h)$.

C Proof of the Misalignment Theorem

Proof of Theorem 6.4. We provide a detailed proof of the misalignment correction.

Step 1: Fee decomposition. The aligned fee sum has N terms:

$$\Psi_{\text{align}} = \frac{\phi}{1-\phi} \sum_{k=-N/2}^{N/2-1} c(h)\, e^{kh/2}\, \mathbb{E}[L_T^{kh}]. \tag{89}$$

Under misalignment with offset δ at the lower boundary and δ' at the upper boundary, the sum modifies the first and last terms:

$$\begin{aligned}\Psi_{\text{mis}} = \frac{\phi}{1-\phi}\Big[&R(\delta)\, c(h)\, e^{x_b h/2}\, \mathbb{E}[L_T^{x_b h}] \\ &+ \sum_{k=\text{interior}} c(h)\, e^{kh/2}\, \mathbb{E}[L_T^{kh}] \\ &+ R(\delta')\, c(h)\, e^{x_t h/2}\, \mathbb{E}[L_T^{x_t h}]\Big]. \end{aligned} \tag{90}$$

Step 2: Ratio. To leading order, all terms in the sum contribute equally (since $\mathbb{E}[L_T^{kh}] \approx \mathbb{E}[L_T^0]$ for kh in the range):

$$\begin{aligned}\frac{\Psi_{\text{mis}}}{\Psi_{\text{align}}} &\approx \frac{(N-2) + R(\delta) + R(\delta')}{N} \\ &= 1 - \frac{(1-R(\delta)) + (1-R(\delta'))}{N}. \end{aligned} \tag{91}$$

Step 3: Substitution. Using $1 - R(\delta) = 1 - 2(1-\delta)/(2-\delta) = \delta/(2-\delta)$:

$$\frac{\Psi_{\text{mis}}}{\Psi_{\text{align}}} = 1 - \frac{1}{N}\Big[\frac{\delta}{2-\delta} + \frac{\delta'}{2-\delta'}\Big] + O(N^{-2}). \tag{92}$$

The $O(N^{-2})$ remainder comes from the variation of $\mathbb{E}[L_T^{kh}]$ across the range (the boundary ticks have slightly different local time values than the center). □

D Second-Order Expansion Details

We collect the detailed computations for the second derivatives of f and g used in Lemma 5.4.

D.1 Second derivative of $\mathbb{E}[L_T^x]$

From (31):

$$\partial_x^2 p_X(t,0,x) = p_X \cdot \Big[\frac{(x-\mu t)^2}{\sigma^4 t^2} - \frac{1}{\sigma^2 t}\Big]. \tag{93}$$

At $x = 0$:

$$\partial_x^2 p_X(t,0,0) = p_X(t,0,0)\Big[\frac{\mu^2}{\sigma^4} - \frac{1}{\sigma^2 t}\Big]. \tag{94}$$

Integrating:

$$\partial_x^2 \mathbb{E}[L_T^x]\big|_0 = \frac{\mu^2}{\sigma^4}\, \mathbb{E}[L_T^0] - \sigma^2 \int_0^T \frac{p_X(t,0,0)}{\sigma^2 t}\, \mathrm{d}t = \frac{\mu^2}{\sigma^4}\, \mathbb{E}[L_T^0] - \lambda_{-1}, \tag{95}$$

where $\lambda_{-1} = \int_0^T p_X(t,0,0)/t\, \mathrm{d}t$ is a divergent integral that cancels in the difference $f''(0) - g''(0)$.

D.2 Second derivative of f

From $f(x) = e^{x/2}\, \mathbb{E}[L_T^x]$:

$$f''(x) = \frac{1}{4}\, e^{x/2}\, \mathbb{E}[L_T^x] + e^{x/2}\, \partial_x \mathbb{E}[L_T^x] + e^{x/2}\, \partial_x^2 \mathbb{E}[L_T^x]. \tag{96}$$

At $x = 0$:

$$f''(0) = \frac{1}{4}\,\lambda_0 + \frac{\mu}{\sigma^2}\,\lambda_0 + \lambda_2, \tag{97}$$

where $\lambda_0 = \mathbb{E}[L_T^0]$ and $\lambda_2 = \partial_x^2 \mathbb{E}[L_T^x]\big|_0$.

Under GBM ($\mu = -\sigma^2/2$): $f''(0) = \lambda_0/4 - \lambda_0/2 + \lambda_2 = -\lambda_0/4 + \lambda_2$.

D.3 Second derivative of g

From $g(x) = e^{-3x/2}\,\mathbb{E}[L_T^x]$:

$$g''(0) = \frac{9}{4}\,\lambda_0 - \frac{3\mu}{\sigma^2}\,\lambda_0 + \lambda_2. \tag{98}$$

Under GBM: $g''(0) = 9\lambda_0/4 + 3\lambda_0/2 + \lambda_2 = 15\lambda_0/4 + \lambda_2$.

D.4 Difference

Under GBM:

$$f''(0) - g''(0) = (-\lambda_0/4 + \lambda_2) - (15\lambda_0/4 + \lambda_2) = -4\lambda_0. \tag{99}$$

Note that λ_2 cancels. This cancellation is essential: the second-order correction $-N^2h^2/6$ is determined entirely by λ_0, not by the more complex quantity λ_2.

For general α:

$$\begin{aligned} f''(0) - g''(0) &= \Big(\frac{1}{4} + \frac{\mu}{\sigma^2}\Big)\lambda_0 - \Big(\frac{9}{4} - \frac{3\mu}{\sigma^2}\Big)\lambda_0 \\ &= \Big(-2 + \frac{4\mu}{\sigma^2}\Big)\lambda_0 = -(2 + 2\alpha)\lambda_0. \end{aligned} \tag{100}$$

Under GBM ($\alpha = 1$): $f''(0) - g''(0) = -4\lambda_0$, as before.

References

H. Adams, N. Zinsmeister, M. Salem, R. Keefer, and D. Robinson. Uniswap v3 core. Technical report, Uniswap Labs, 2021.

M. Broadie, P. Glasserman, and S. Kou. A continuity correction for discrete barrier options. *Math. Finance*, 7(4):325–349, 1997.

M. Broadie, P. Glasserman, and S. Kou. Connecting discrete and continuous path-dependent options. *Finance Stoch.*, 3:55–82, 1999.

A. Carta and S. Fabrizio. Fee revenue for concentrated liquidity providers in decentralized exchanges. *Quant. Finance*, 24(6):789–805, 2024.

A. Clark and B. K. Hassani. Liquidity provider returns in geometric mean markets. *Digital Finance*, 5:9–45, 2023.

M. Echenim, E. Gobet, and A.-C. Maurice. Uniswap v3: impermanent loss modeling and swap fees asymptotic analysis. *SIAM J. Financial Math.*, 16(2):310–349, 2025.

L. Feng and V. Linetsky. Pricing discretely monitored barrier options and defaultable bonds in Lévy process models: a fast Hilbert transform approach. *Math. Finance*, 18(3):337–384, 2008.

M. Fukasawa, B. Majó, and M. Worhacz. Optimal liquidity provision in concentrated liquidity markets. Working paper, 2023.

E. Gobet and N. Landon. Almost sure optimal hedging strategy. *Ann. Appl. Probab.*, 24(4):1652–1690, 2014.

E. Gobet and U. Stazhynski. Parametric inference for diffusions observed at stopping times. *Electron. J. Statist.*, 14(1):2098–2122, 2020.

P. Hörfelt. Extension of the corrected barrier approximation by Broadie, Glasserman, and Kou. *Finance Stoch.*, 7(2):231–243, 2003.

S. Howison and M. Steinberg. A matched asymptotic expansions approach to continuity corrections for discretely sampled options. Part 1: barrier options. *Appl. Math. Finance*, 14(1):63–89, 2007.

S. Loesch, N. Hindman, M. B. Richardson, and N. Welch. Impermanent loss in Uniswap v3. Working paper, 2021.

J. Milionis, C. Moallemi, T. Roughgarden, and A. L. Zhang. Automated market making and loss-versus-rebalancing. Working paper, Columbia University, 2022.

A. Paone. Designing barrier options using concentrated liquidity of constant function market makers. SSRN 5233590, 2025.